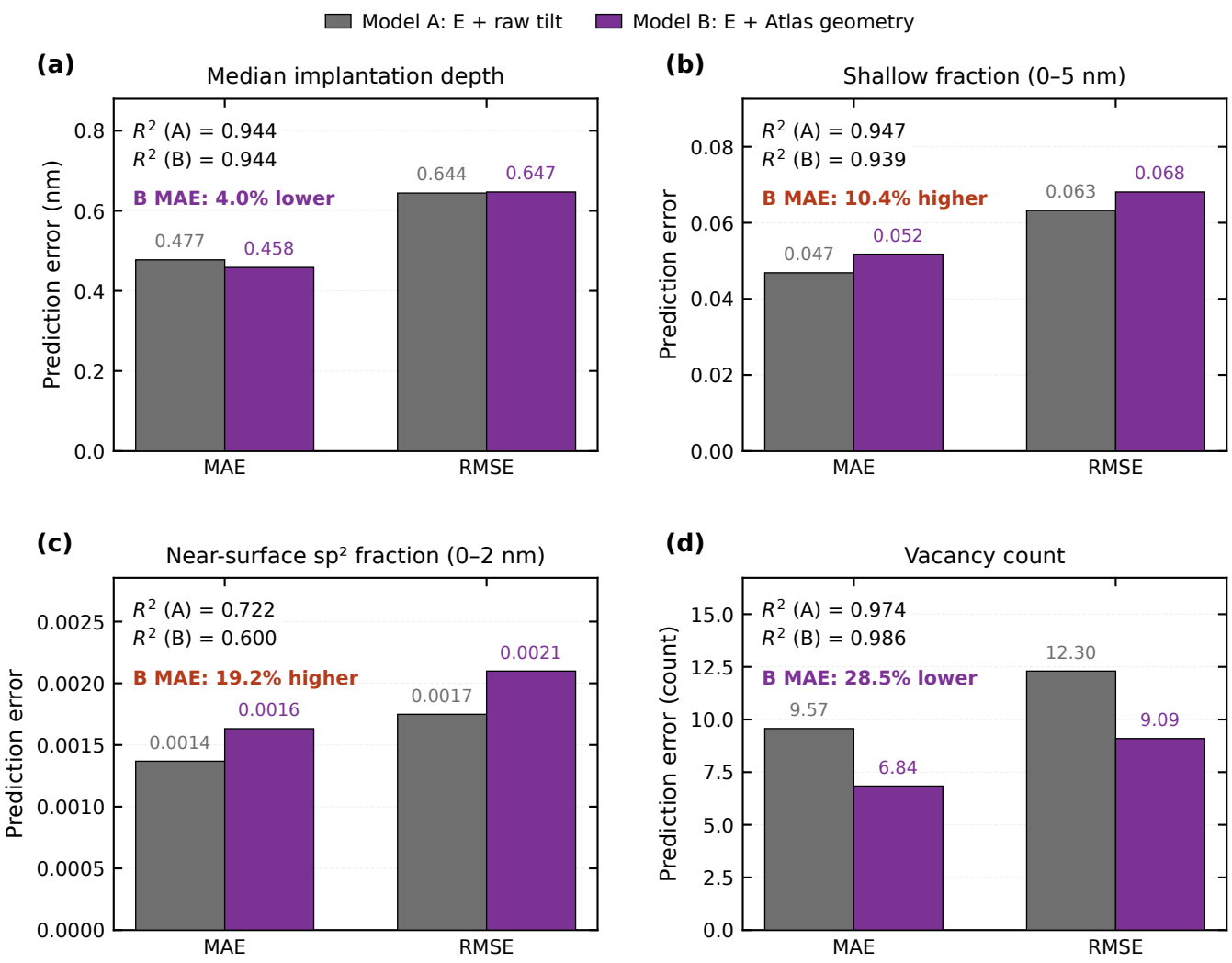
Model A: E + raw tilt
Model B: E + Atlas geometry
(a)
Median implantation depth
R^2 (A) = 0.944
R^2 (B) = 0.944
B MAE: 4.0% lower
Prediction error (nm)
0.477
0.458
0.644
0.647
MAE
RMSE
(b)
Shallow fraction (0–5 nm)
R^2 (A) = 0.947
R^2 (B) = 0.939
B MAE: 10.4% higher
Prediction error
0.047
0.052
0.063
0.068
MAE
RMSE
(c)
Near-surface sp² fraction (0–2 nm)
R^2 (A) = 0.722
R^2 (B) = 0.600
B MAE: 19.2% higher
Prediction error
0.0014
0.0016
0.0017
0.0021
MAE
RMSE
(d)
Vacancy count
R^2 (A) = 0.974
R^2 (B) = 0.986
B MAE: 28.5% lower
Prediction error (count)
9.57
6.84
12.30
9.09
MAE
RMSE